

# Chess math: count the whole exchange

Chess Math

## THE IDEA

Use material arithmetic to compare complete sequences. Assign approximate values  $P=1$ ,  $N=3$ ,  $B=3$ ,  $R=5$ ,  $Q=9$ , then subtract everything you give up from everything you gain. Stop only when immediate recaptures have been considered.

## WORKED EXAMPLE

You give a rook and a pawn for a bishop and a knight. The loss is  $5+1=6$ ; the gain is  $3+3=6$ . Material is roughly equal, but two active minor pieces may coordinate differently from a rook in an open position.

## TRY IT BEFORE READING ON

**What is the nominal balance if you win a rook and lose a bishop?**

## ANSWER AND REASONING

$5-3=2$  pawn units in your favor. That is winning the exchange. Check for extra pawn losses and king danger before treating the two-point gain as decisive.

## TAKE THIS INTO YOUR NEXT GAME

**Arithmetic evaluates a sequence, not an isolated capture.**